

Normal distribution (Gaussian)

$$\text{pdf } f(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

$$N(\mu, \sigma) \quad \mu, \sigma - \text{parameters}$$

$$-\infty < x < \infty, \quad \sigma > 0$$

$$\mu=0, \sigma=1 \quad f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

$N(0,1)$ - standard normal distribution

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx \quad \left| \quad E(X-\mu) \right.$$

$$E(X-\mu) = \int_{-\infty}^{\infty} (x-\mu) \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sqrt{2\pi} \sigma} \int_{-\infty}^{\infty} z e^{-\frac{z^2}{2\sigma^2}} dz$$

$$= \frac{1}{\sqrt{2\pi} \sigma} \times 0$$

$$x-\mu = z$$

$$dx = dz$$

$$\frac{z^2}{2\sigma^2} = t$$

$$\frac{z dz}{\sigma^2} = dt$$

$$E(X) = \mu.$$

$$\text{Var}(X) = \sigma^2$$

$$E(X-\mu)^2 = \sigma^2 \quad \text{— check}$$

$$E(X-\mu)^2 = \frac{1}{\sqrt{2\pi} \sigma} \int_{-\infty}^{\infty} (x-\mu)^2 e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} dx$$

$$= \frac{1}{\sqrt{2\pi} \sigma} \int_{-\infty}^{\infty} z^2 e^{-\frac{1}{2} \frac{z^2}{\sigma^2}} dz \quad \begin{matrix} (x-\mu) = z \\ dx = dz \end{matrix}$$

$$= \frac{1}{\sqrt{2\pi} \sigma} \times 2 \int_0^{\infty} z^2 e^{-\frac{z^2}{2\sigma^2}} dz$$

$$= \frac{1}{\sqrt{2\pi} \sigma} \times 2 \cdot \int_0^{\infty} z^2 e^{-t} \frac{dz}{\sqrt{2\sigma^2 t}}$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \int_0^{\infty} t^{\frac{1}{2}} e^{-t} dt$$

$$\frac{z^2}{2\sigma^2} = t$$

$$\frac{z dz}{\sigma^2} = dt$$

$$z dz = \sigma^2 dt$$

$$\textcircled{dz} = \frac{\sigma^2 dt}{\sqrt{2\sigma^2 t}}$$

function

Gamma function

$$\Gamma a = \int_0^{\infty} x^{a-1} e^{-x} dx; \quad a > 0$$

 $a = n \rightarrow \text{integer}$

$$\Gamma n = (n-1)!$$

$$\Gamma \frac{1}{2} = \sqrt{\pi}$$

$$E(X - \mu)^2 = \frac{2\sigma^2}{\sqrt{\pi}} \int_0^{\infty} t^{\frac{1}{2}} e^{-t} dt$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \int_0^{\infty} t^{\frac{3}{2}-1} e^{-t} dt$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \Gamma \frac{3}{2}$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2}$$

$$= \sigma^2$$

$$\Gamma a = (a-1)\Gamma(a-1)$$

1

 μ : median.

Me is the median of a cont rv X
 $P(X \geq Me) = \frac{1}{2}$ $P(X \leq Me) = \frac{1}{2}$

$$\Leftrightarrow \underline{P(X \leq Me) = \frac{1}{2}}$$

$$\text{Median} = \inf \left\{ x: F(x) \geq \frac{1}{2} \right\}$$

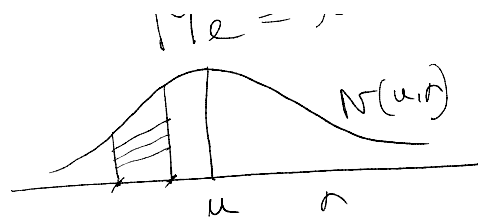
$$\S P(X \leq Me) = \frac{1}{2}$$

$$\int_{-\infty}^{Me} f(x) dx = \frac{1}{2}$$

$$Me = \mu.$$


 $\mu(u, \sigma)$

$$P(a \leq X \leq b)$$

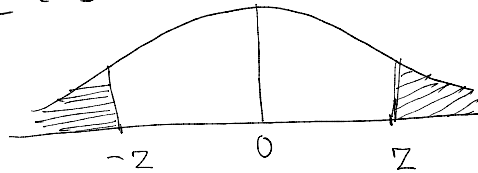


$$P(a \leq X \leq b)$$

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$$\Phi(-z) = 1 - \Phi(z), \quad \Phi(0) = \frac{1}{2}$$

$z > 0$



$$P(a \leq X < b)$$

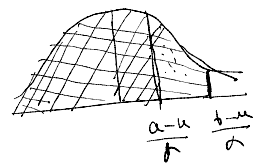
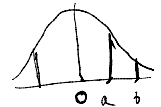
$$(X \sim N(u, \sigma) \Rightarrow \frac{X-u}{\sigma} \sim N(0, 1))$$

$$= P\left(\frac{a-u}{\sigma} \leq \frac{X-u}{\sigma} \leq \frac{b-u}{\sigma}\right)$$

$$Z \sim N(0, 1)$$

$a < b$

$$= \Phi\left(\frac{b-u}{\sigma}\right) - \Phi\left(\frac{a-u}{\sigma}\right)$$



Central moments.

$$\mu_{2r+1} = 0$$

$$\mu_{2r} = \sigma^{2r} \frac{(2r)!}{2^r (r!)}$$

$$\frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} (x-u)^{2r+1} e^{-\frac{1}{2} \frac{(x-u)^2}{\sigma^2}} dx$$